Mufeng Chen

Ph.D. Student, Electrical and Computer Engineering

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Summary

- Ph.D. student at Purdue ECE working on memory-centric architecture and circuits: analog hybrid memory systems fusing eDRAM, RRAM and BEOL oxide-semiconductor FETs, and High-Bandwidth Flash as a new tier in the memory hierarchy.

Education

PhD — **Purdue University**, Electrical and Computer Engineering — West Lafayette, IN, USA — 2024 – present
- Advised by Prof. Haitong Li in the NanoX Lab.
- Research focus — analog hybrid memory systems, and High-Bandwidth Flash as a new memory tier.

BS — **Huazhong University of Science and Technology**, Optics and Electronic Engineering — Wuhan, China — 2017 – 2021

Experience

Purdue University, Graduate Research Assistant, NanoX Lab — West Lafayette, IN, USA — 2024 – present — 2 years
Advised by Prof. Haitong Li.
- Analog hybrid memory macros fusing eDRAM, RRAM and BEOL oxide-semiconductor FETs.
- Zeroth-order optimization for on-device LLM fine-tuning.
- High-Bandwidth Flash as a memory tier: accelerator interface, placement, and cross-accelerator pooling.

Zhejiang University, Research Assistant, RFNE Research Center — Hangzhou, China — 2021 – 2024 — 3 years
Advised by Prof. Er-Ping Li. Group leader of the RRAM-eDRAM hybrid CiM design program.
- Group leader for the RRAM-eDRAM hybrid CiM design program.
- SNN-based vision models, multimodal learning in SNNs, hybrid DNN-HDC.
- System-level specification, digital design, and heterogeneous integration of fused CiM.

Purdue University, Research Intern, NanoX Lab — West Lafayette, IN, USA — 2023 – 2023 — 1 year
Summer internship, advised by Prof. Haitong Li.
- Monolithic 3D-RRAM compute-in-memory.
- Hyperdimensional computing for vision transformers.

Rice University, Research Intern, SIMS Lab — Houston, TX, USA — 2022 – 2022 — 1 year
Summer internship, advised by Prof. Kaiyuan Yang.
- Charge-domain SRAM-CiM for Bayesian neural networks.

Wuhan National Laboratory for Optoelectronics, Research Assistant — Wuhan, China — 2019 – 2019 — 1 year
Advised by Prof. Chen Lin.
- Optical metasurface design (achromatic metalenses).

Technical Skills

Circuit & EDA: Cadence Virtuoso, Synopsys VCS, Design Compiler, Innovus

Programming: Python (PyTorch), C/C++, MATLAB, Verilog/SystemVerilog

Research areas: Memory hierarchy / HBF, compute-in-memory, hybrid memory systems, mixed-signal CMOS+X, hyperdimensional computing

Transcripts

- B.S. transcript: download
- Graduate transcript: download